Prior Design Attempts and Analysis:

Prior Design Attempt #1 - Davit Crane

Details:

The Davit can support any object under the weight limit of 1000lbs, and it has a crank to allow a human to lift the heavy objects with their hands. It can rotate 360* allowing it to fit a variety of scenarios. It also has 4 height settings further increasing the versatility of the crane. It costs $2500. It also has a small profile of less than 4ft by 2ft.

Analysis:

One limitation of this design is the height the load can be raised. Since the crank is at the top of the vertical pole, the top of the crane has to be within reach of the zookeepers so it can be easily operated. Another potential issue is the length of the arm as the toys have to hang far enough away from the fence to prevent the eland's horns from getting stuck in the chainlink. The design also costs an excessive amount. The design also requires a concrete block to be installed because the screws require an object to be anchored into.

Prior Design Attempt #2 - Standing Basketball Hoop

Details:

This basketball hoop can be lowered to the ground so the toys can be attached, then it can be raised and moved into position over the fence. The hoop can be lifted to 10 feet above the ground, which clears the 8 foot fence. The top beam is 8 feet long so it will reach into the enclosure and hold the toys away from the fence. The main issue with this hoop is its price of nearly $16,000. It is also on wheels so if the eland's horns got caught on the toy, they would be able to pull/move the whole system. This does not have a built in way to attach the toys, so we would have to design and build an addition that would allow the toys to be attached and raised and lowered.

Analysis:

This design would allow the toys to be easily raised and lowered. However, since it does not rotate, it would have to be manually moved into and out of position over the fence each time the toys are changed. Since it's on wheels, the hoop needs a stable platform to be placed on and it could potentially be moved by the elands if they get their horns caught in the toy.

Prior Design Attempt #3 - RESTROOM STALL BLOCK-AIDE

Details:

This punching bag holder can range from 63” to 80” in height, while holding any punching bag or object that can hang from the carabiner up to 132lbs. The triangle base makes it have a profile of 36” by 72” by 88”, but allows it to reach the 132lbs weight limit. It only costs $125, and doesn’t allow movement of the hanging object vertically or horizontally.

Analysis:

The hanger's triangle base solves the issue of weight requirement that our design needs to reach, but this design doesn't address the problem of moving the payload horizontally and vertically. The cost could most likely be reduced further as well. The design also doesn't reach high enough or far enough forward to be used to position the toy in the needed position, but making a larger scaled version of the punching bag hanger could fix that problem.

Citations

1. *"Porter 1135 Competition Manual Portable Basketball Hoop w/ 8' Boom." Pro Sports Equip*, https://prosportsequip.com/products/porter-1135-competition-manual-portable-basketball-hoop-w-8-boom-1135080. Accessed 27 Feb. 2026.

2. *"Soozier Height Adjustable Heavy Punching Bag Hanger Bracket with Heavy Duty Foldable Steel Frame & Weightable Base, Stand Only." Target*, sold and shipped by Aosom, www.target.com/p/soozier-height-adjustable-heavy-punching-bag-hanger-bracket-with-heavy-duty-foldable-steel-frame-weightable-base-stand-only/-/A-90247692. Accessed 27 Feb. 2026.

3. *"Steel Davit Crane | OZ Lifting Products." OZ Lifting Products*, https://www.ozliftingproducts.com/product-category/davit-cranes/steel-davit-cranes/